

Hugging Face Transformers (transformers, pipeline) – for translation and sentiment analysis using pretrained models.

ROUGE & BLEU evaluation (rouge_score, evaluate) – to measure translation quality; **CSV** module – to export results to a file.

Overview of What This Script Does

This script:

1. **Translates text** from various languages to English using an AI model (NLLB).
2. **Detects sentiment** (positive/negative) of the English translation.
3. **Translates** the sentiment result **back into the original language**.
4. **Prints all the results** clearly.
5. **Evaluates** the translation quality using **ROUGE** and **BLEU** scores.



```
from transformers import AutoModelForSeq2SeqLM, AutoTokenizer, pipeline
from rouge_score import rouge_scorer
import evaluate

# -----
# NLLB Language Codes
# -----
lang_code_map = {
    "en": "eng_Latn",
    "fr": "fra_Latn",
    "es": "spa_Latn",
    "ja": "jpn_Jpan",
    "de": "deu_Latn"
}

# -----
# Translation using NLLB
# -----
model_name = "facebook/nllb-200-distilled-600M"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

def translate_text(text, src_lang, tgt_lang="en"):
    src_lang_code = lang_code_map[src_lang]
    tgt_lang_code = lang_code_map[tgt_lang]

    tokenizer.src_lang = src_lang_code
    encoded = tokenizer(text, return_tensors="pt", padding=True)
    generated_tokens = model.generate(**encoded, forced_bos_token_id=tokenizer.convert_tokens_to_ids(tgt_lang_code))
    return tokenizer.decode(generated_tokens[0], skip_special_tokens=True)

# -----
# Sentiment Analysis pipeline
# -----
sentiment_pipeline = pipeline("sentiment-analysis")

def classify_sentiment(text):
    result = sentiment_pipeline(text)[0]
    return result["label"]

# -----
# Input Texts
# -----
```

```

# -----
# Input Texts
# -----
texts = [
    {"text": "Le produit est excellent et la livraison a été rapide.", "lang": "fr"},
    {"text": "El servicio fue terrible y el producto estaba roto.", "lang": "es"},
    {"text": "この商品はとても良いです。ありがとう。", "lang": "ja"},
    {"text": "Das war eine schlechte Erfahrung mit dem Kundenservice.", "lang": "de"},
    {"text": "The quality exceeded expectations and the packaging was great.", "lang": "en"},
]

# -----
# Translation, Sentiment, and Reverse Translation
# -----
annotations = []
for item in texts:
    original_text = item["text"]
    lang = item["lang"]

    # Translate to English if needed
    if lang != "en":
        translated = translate_text(original_text, src_lang=lang, tgt_lang="en")
    else:
        translated = original_text

    # Run sentiment classification
    sentiment = classify_sentiment(translated)

    # Translate sentiment back to original language
    if lang != "en":
        sentiment_localized = translate_text(sentiment, src_lang="en", tgt_lang=lang)
    else:
        sentiment_localized = sentiment

    annotations.append({
        "original_text": original_text,
        "translated": translated,
        "sentiment_en": sentiment,
        "sentiment_localized": sentiment_localized
    })

# -----
# Output Results
# -----

```

```

# -----
# Output Results
# -----
for a in annotations:
    print("\n---")
    print(f"Original: {a['original_text']}")
    print(f"Translated: {a['translated']}")
    print(f"Sentiment (EN): {a['sentiment_en']}")
    print(f"Sentiment (Localized): {a['sentiment_localized']}")

# -----
# Evaluation with ROUGE & BLEU
# -----
candidate = "The product is excellent and delivery was fast."
reference = "The product is great and arrived quickly."

# ROUGE
scorer = rouge_scorer.RougeScorer(['rouge1', 'rougeL'], use_stemmer=True)
rouge_scores = scorer.score(reference, candidate)
print("\nROUGE Scores:", rouge_scores)

# BLEU (✅ Correct format)
bleu = evaluate.load("bleu")
bleu_score = bleu.compute(
    predictions=[candidate],          # List of strings
    references=[[reference]]          # List of list of strings
)
print("BLEU Score:", bleu_score)

```

```

ubuntu@ubuntu: ~/LLM
(myenv) ubuntu@ubuntu:~/LLM$ python3 LLM_translator_3.py
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f (https://huggingface.
distilbert/distilbert-base-uncased-finetuned-sst-2-english).
Using a pipeline without specifying a model name and revision in production is not recommended.
Device set to use cpu

---
Original: Le produit est excellent et la livraison a été rapide.
Translated: The product is excellent and the delivery was quick.
Sentiment (EN): POSITIVE
Sentiment (Localized): C'est positif

---
Original: El servicio fue terrible y el producto estaba roto.
Translated: The service was terrible and the product was broken.
Sentiment (EN): NEGATIVE
Sentiment (Localized): - No .

---
Original: この商品はとても良いです。ありがとう。
Translated: This product is very good. Thank you.
Sentiment (EN): POSITIVE
Sentiment (Localized): ポジティブ

---
Original: Das war eine schlechte Erfahrung mit dem Kundenservice.
Translated: That was a bad experience with customer service.
Sentiment (EN): NEGATIVE
Sentiment (Localized): - Nein

---
Original: The quality exceeded expectations and the packaging was great.
Translated: The quality exceeded expectations and the packaging was great.
Sentiment (EN): POSITIVE
Sentiment (Localized): POSITIVE

```

```
---
Original: この商品はとても良いです。ありがとう。
Translated: This product is very good. Thank you.
Sentiment (EN): POSITIVE
Sentiment (Localized): ポジティブ

---
Original: Das war eine schlechte Erfahrung mit dem Kundenservice.
Translated: That was a bad experience with customer service.
Sentiment (EN): NEGATIVE
Sentiment (Localized): - Nein

---
Original: The quality exceeded expectations and the packaging was great.
Translated: The quality exceeded expectations and the packaging was great.
Sentiment (EN): POSITIVE
Sentiment (Localized): POSITIVE

ROUGE Scores: {'rouge1': Score(precision=0.5, recall=0.5714285714285714, fmeasure=0.5333333333333333), 'rougeL': Score(precision=0.5, recall=0.5714285714285714, fmeasure=0.5333333333333333)}
BLEU Score: {'bleu': 0.0, 'precisions': [0.5555555555555556, 0.25, 0.14285714285714285, 0.0], 'brevity_penalty': 1.0, 'length_ratio': 1.125, 'translation_length': 9, 'reference_length': 8}
(myenv) ubuntu@ubuntu:~/LLM$
```